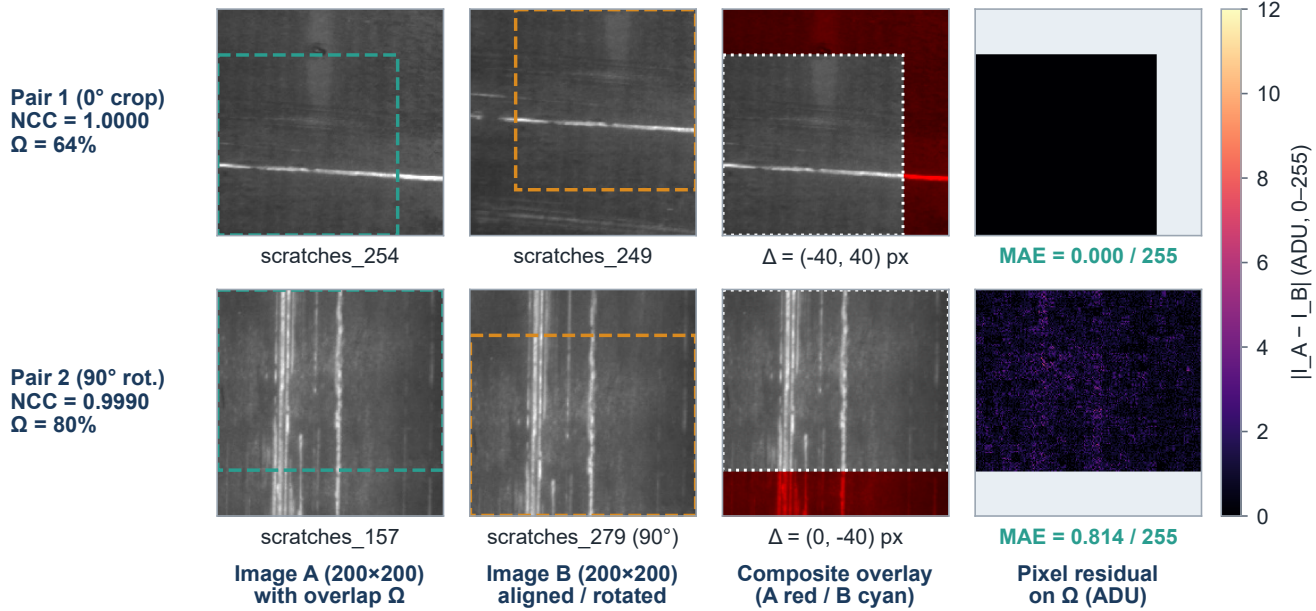

Pair 1 (0° crop)
NCC = 1.0000
Ω = 64%
scratches_254
scratches_249
Δ = (-40, 40) px
MAE = 0.000 / 255
Pair 2 (90° rot.)
NCC = 0.9990
Ω = 80%
scratches_157
scratches_279 (90°)
Δ = (0, -40) px
MAE = 0.814 / 255
Image A (200×200) with overlap Ω
Image B (200×200) aligned / rotated
Composite overlay (A red / B cyan)
Pixel residual on Ω (ADU)
|I_A − I_B| (ADU, 0–255)
12
10
8
6
4
2
0